



SETS

CMSC 56 – DISCRETE MATHEMATICS IN COMPUTER SCIENCE I

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Major Reference Books:

Kenneth H. Rosen (2012). Discrete Mathematics and Its Applications. 7th Edition. The McGraw-Hill Companies, Inc.

K.H. Rosen, J.G. Michaels, J.L. Gross, J.W. Grossman, D.R. Shier (2000). Handbook of Discrete and Combinatorial Mathematics, CRC Press LLC



At the end of this lecture, the students should be able to:

- Define what is a set
- Classify the different types of set
- Demonstrate the different relations and operations involving sets
- Apply the properties of set operations in real life problems using Venn diagrams
- Prove equality of sets using Venn diagrams
- Identify whether a set of subsets is a partition or not



What is a set?

A **set** is a collection of well-defined objects.

What is a set?

An object may or may not belong to a set.

Objects that belong to a set are called elements of that particular set.

Notation for element: $\in$

Notation for not an element: $\notin$

Describing a set

A set could be described by two common methods:

1. **Roster (list) Method**
2. **Rule Method**

Roster Method

In the roster method, the elements are enumerated, separated by **commas** and enclosed by **braces**.

Important note: The elements must be distinct!

Rule Method

In the rule method, the elements are **not** listed individually.

Instead, membership in the set is defined by stating a descriptive phrase that gives the common property of the elements.

Rule Method

Note: There are sets which could not be described using the rule method.

Remember: The rule method can be used as long as there is a property that will describe each element of the set.

Types of Sets

1. Empty set
2. Finite and infinite sets
3. Equivalent Sets
4. Subset
5. Proper subset
6. Equal sets
7. Universal Set
8. Power set
9. Cartesian Product of sets A and B

1. Empty set

Sets having no elements are called **empty** or **null sets**.

An empty set is denoted by $\emptyset$ or $\{\}$.

2. Finite and infinite sets

According to the size of sets, a set can be classified as either finite or infinite.

A set is said to be **finite** if it is possible to write down a complete list of all elements of the set; otherwise the set is said to be **infinite**.

Cardinality

The number of elements of a finite set is called the **cardinality** of the set.

Notation: $|A|$ (Cardinality of set A)

3. Equivalent sets

Sets A and B are equivalent sets if and only if there exists a one-to-one correspondence between sets A and B .

If A and B are equivalent, we write

$$A \approx B$$

4. Subset

Set A is a subset of B if and only if each element of A is a member of B .

If A is a subset of B , we write $A \subseteq B$.

5. Proper Subset

If A is a subset of B and there is at least one element of B which is not an element of A , then A is a **proper subset** of B .

If A is a proper subset of B , we write $A \subset B$.

6. Equal Sets

Two sets A and B having the same elements are said to be **equal**.

If A is equal to B , we write $A = B$.

Observe that $A = B$ if and only if A is a subset of B and B is a subset of A .

7. Power Set

The set whose elements are the subsets of a set A is called the **power set of A** . denoted by P^A or $\mathcal{P}(A)$.

Notation: P^A or $\mathcal{P}(A)$.

8. Universal Set

A fixed set that contains all sets under discussion is called a universal set.

Notation: U

Exercise

Classify whether the set is

- a) written in Roster or Rule method
- b) finite or infinite
- c) subset or not a subset of the set of integers

1. $W = \{0, 1, 2, 3, \dots, 1\,000\,000\}$

2. $Y = \{x, y, z, a, b, c\}$

3. $D =$
 $\{d \mid d \text{ is a real number in the English alphabet}\}$

4. $N = \{n \mid n \text{ is a real number}\}$

Answer to the Exercise

Classify whether the set is

- a) written in Roster or Rule method
- b) finite or infinite
- c) subset or not a subset of the set of integers

1. $W = \{0, 1, 2, 3, \dots, 1\,000\,000\}$ - Roster, finite, yes
2. $Y = \{x, y, z, a, b, c\}$ - Roster, finite, no
3. $D = \{d \mid d \text{ is a letter the English alphabet}\}$
- Rule, finite, no
4. $N = \{n \mid n \text{ is a real number}\}$ - Rule, infinite, no

9. Cartesian Product

Let A and B be sets. **The Cartesian product of A and B** , denoted by

$$A \times B$$

is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

9. Cartesian Product

NOTE: The Cartesian Products

$A \times B$ and $B \times A$ are not equal,
unless $A = \emptyset$ or $B = \emptyset$ or $A = B$.

Exercise

Give the Cartesian Products

a) $A \times B$

b) $B \times A$

where

$$A = \{0, 1, 2, 3\}$$

$$B = \{x, y, z\}$$

Give the cardinality of the two sets.

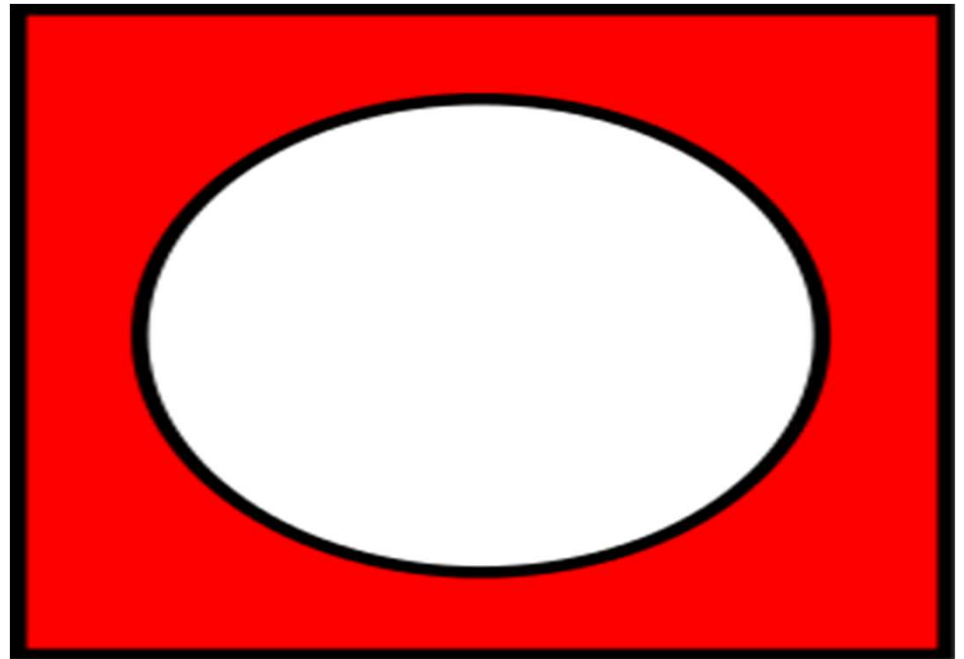
Operations on sets

1. Complement
2. Set difference
3. Set union
4. Set intersection
5. Other properties

1. Complement

The complement of a set A is the set of elements that is in the universal set which are **not** in A .

Notation: A' or A^c



1. Complement

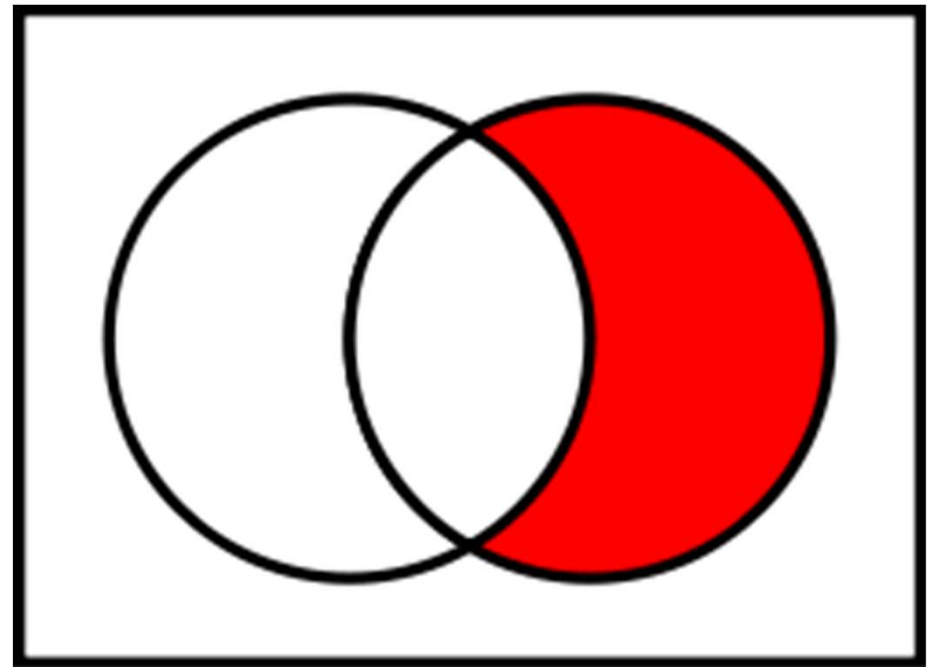
Some properties of the complement:

- $\emptyset^C = U$
- $U^C = \emptyset$
- $(A^C)^C = A$

2. Set difference

The set difference of B and A , also called “ B but not A ”, is the set of elements in B which are not elements of A .

Notation: $B - A$



2. Set difference

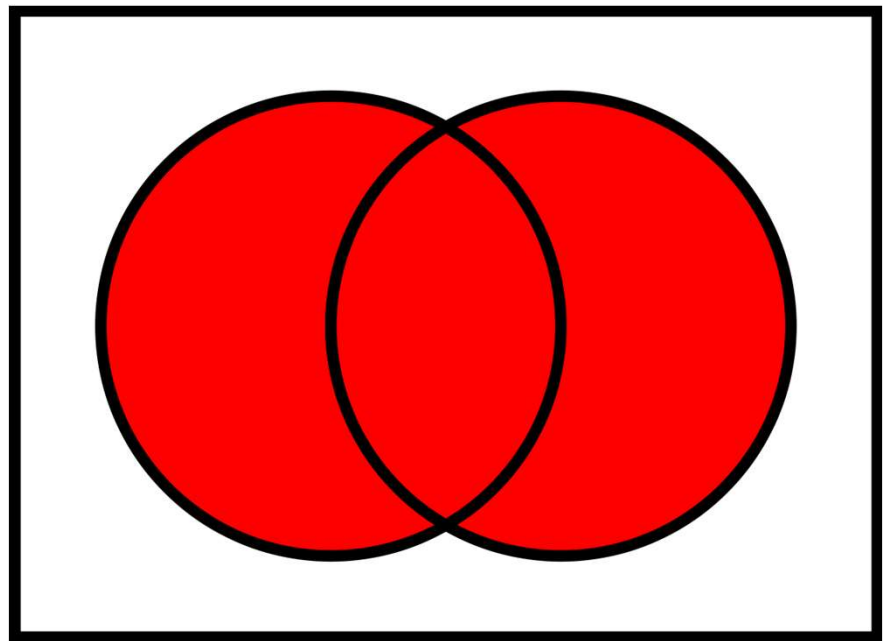
Some properties of the set difference:

- $U - A = A^c$
- $A - \emptyset = A$
- $A - A = \emptyset$

3. Set Union

The union of sets A and B is the set of elements in A or in B .

Notation: $A \cup B$



3. Set union

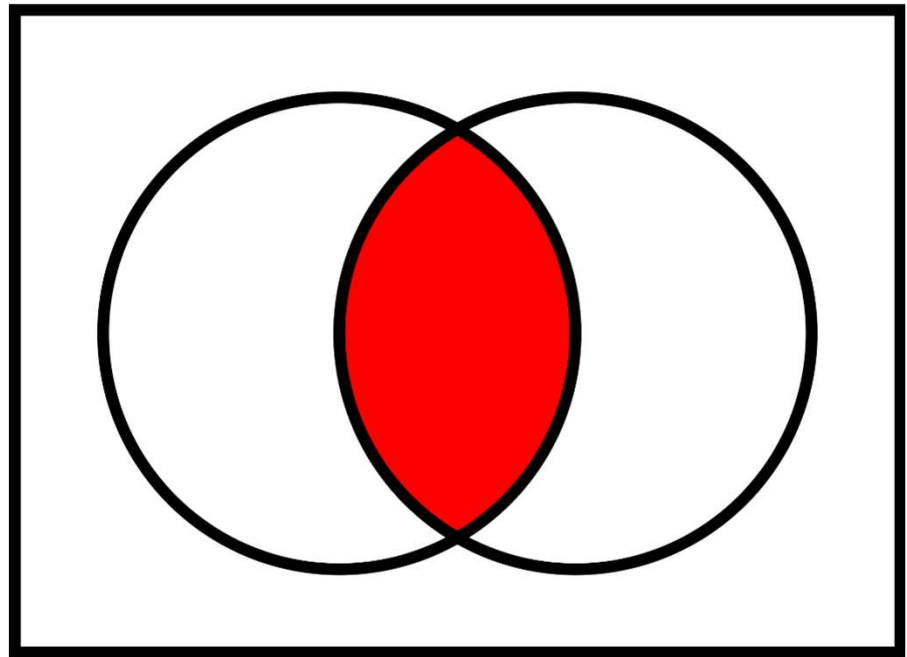
Some properties of the set union:

- $A \cup B = B \cup A$
- $(A \cup B) \cup C = A \cup (B \cup C)$

4. Set intersection

The intersection of sets A and B is the set of elements that are common to both A or B .

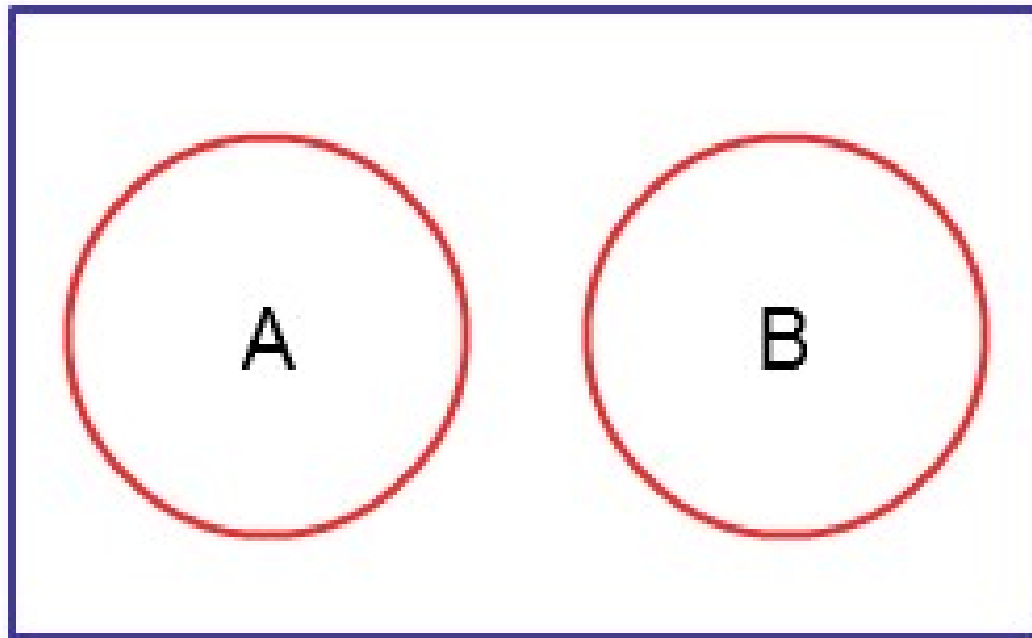
Notation: $A \cap B$



4. Set intersection

Two sets A and B are **disjoint** if:

$$A \cap B = \emptyset$$



5. Other properties

- Distributive Property
- De Morgan's Law
- Double complement
- Idempotent Laws
- Identity Laws
- Inverse Laws
- Domination Laws
- Absorption Laws

5. Other properties

- Distributive Property

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- De Morgan's Law

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

- Double complement

$$(A^c)^c = A$$

5. Other properties

- Idempotent Laws

$$A \cup A = A$$

$$A \cap A = A$$

- Identity Laws

$$A \cup \emptyset = A$$

$$A \cap U = A$$

- Inverse Laws

$$A \cup A^c = U$$

$$A \cap A^c = \emptyset$$

5. Other properties

- Domination Laws

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

- Absorption Laws

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Summary of Set Properties / Tables of Laws

Commutative Laws	
(1) $A \cup B = B \cup A$	(1') $A \cap B = B \cap A$
Associative Laws	
(2) $A \cup (B \cup C) = (A \cup B) \cup C$	(2') $A \cap (B \cap C) = (A \cap B) \cap C$
Distributive Laws	
(3) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	(3') $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity Laws	
(4) $A \cup \emptyset = \emptyset \cup A = A$	(4') $A \cap U = U \cap A = A$
Complement Laws	
(5) $A \cup A^c = U$	(5') $A \cap A^c = \emptyset$
Idempotent Laws	
(6) $A \cup A = A$	(6') $A \cap A = A$
Null Laws	
(7) $A \cup U = U$	(7') $A \cap \emptyset = \emptyset$
Absorption Laws	
(8) $A \cup (A \cap B) = A$	(8') $A \cap (A \cup B) = A$
DeMorgan's Laws	
(9) $(A \cup B)^c = A^c \cap B^c$	(9') $(A \cap B)^c = A^c \cup B^c$
Involution Law	
(10) $(A^c)^c = A$	

Venn Diagram

A pictorial representation of sets is called a **Venn diagram**. (You have seen pictures when we discussed set operations. Those are called Venn diagrams.)

- The universal set is represented by the interior of a rectangle.
- The subsets of U is represented by any closed plane figures like circle, triangle, or square

Proving with Laws of Sets

There are many ways to prove a theorem on sets. The following are some of them:

1. **Proof using previously proven theorems**
2. **Proof using the Indirect Method/Contradiction** – the technique here is to assume that the conclusion is false and obtain a contradiction.
3. **Proof using Venn Diagrams** – the technique here is to illustrate both sides of the statement using Venn diagrams and determine whether both diagrams give the same drawings.
4. **Proof using Set-membership Tables** – here, we let A be a subset of a universal set U and let $u \in U$. To use this method, we note that exactly one of the following is true: a) $u \in A$ (*write 1 in the table*) or b) $u \notin A$ (*write 0 in the table*).
5. **Disproof by Counterexample** – the technique is to disprove a theorem by finding an example that violates the theorem.

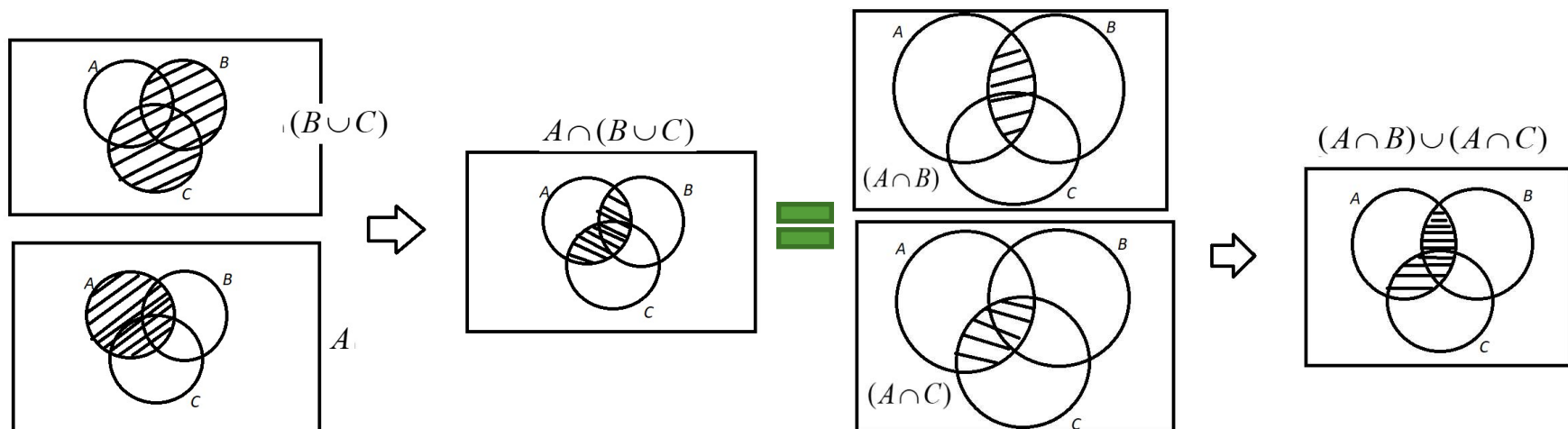
3. Proof using Venn Diagrams

Example:

Prove one of the distributive law:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Proof:



3. Proof using Venn Diagrams

Exercise:

Let A and B be sets. Then

$$A - B = A \cap B^c$$

Some Theorems on Sets

Let $A, B, C \subseteq U$

- a) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- b) If $A \subset B$ and $B \subseteq C$, then $A \subset C$.
- c) If $A \subseteq B$ and $B \subset C$, then $A \subset C$.
- d) If $A \subset B$ and $B \subset C$, then $A \subset C$.

Some Theorems on Sets

For any universe U , let $A \subseteq U$. Then

$$\emptyset \subseteq A$$

and if $A \neq \emptyset$, then

$$\emptyset \subset A.$$

Some Theorems on Sets

For any universe U , and sets $A, B \subseteq U$, the following statements are equivalent:

- a) $A \subseteq B$
- b) $A \cap B = A$
- c) $A \cup B = B$
- d) $B^c \subseteq A^c$

More About Equal Sets

To show that two sets A and B are equal, you can use the following methods:

- Double inclusion: $A \subseteq B$ and $B \subseteq A$
- Set Identities
- Membership tables

More About Equal Sets

Exercise:

1. Show that

$$(B - A) \cup (C - A) = (B \cup C) - A$$

2. Show that $(A - B) - (B - C) = A - B$

3. Show that

$$[(A - B) - (B - C)]' = A' \cup B$$

More about Set Operations

The set $\{ x \mid \text{some property of } x \}$ is the set of all objects x that satisfy the given property. Formally we have that $w \in \{ x \mid \text{some property of } x \}$ if and only if the specified property holds for w .

The set $A \cup B$ is the set $\{ x \mid x \in A \text{ or } x \in B \}$. Equivalently, $x \in A \cup B$ if and only if $x \in A$ or $x \in B$.

The set $A \cap B$ is the set $\{ x \mid x \in A \text{ and } x \in B \}$. Equivalently, $x \in A \cap B$ precisely if $x \in A$ and $x \in B$.

The set $A - B$ is the set $\{ x \mid x \in A \text{ and } x \notin B \}$. This set is also sometimes denoted $A \setminus B$.

The set $A \Delta B$ is the set $\{ x \mid \text{exactly one of } x \in A \text{ and } x \in B \text{ is true} \}$.

Partition of a Set

A **partition of a set** is a grouping of the set's elements into non-empty subsets, in such a way that every element is included in one and only one of the subsets

Partition of a Set

A collection of nonempty sets $\{A_1, A_2, \dots, A_n\}$ is a partition of a set A if and only if

- $A = A_1 \cup A_2 \cup \dots \cup A_n$
- $A_1, A_2, \dots, A_n$ are mutually disjoint



Partition of a Set

Exercise:

The set of even and odd integers